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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Café Sales Data Cleaning Using Power Query

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Café Sales Data Cleaning Using Power Query

Github Link: <https://github.com/Bhumikaravi/Business-Analysis-Projects>

Problem Statement

A retail company receives sales data from multiple sources: sales data stored in Excel files, customer data stored in CSV files, and product information stored in another source such as a secondary Excel file or database. This data contains missing values, duplicate records, and inconsistent entries. Management requires a centralized reporting system, so the datasets must be imported into Power BI, appropriately connected, cleaned, and prepared for reliable business reporting.

Steps followed

Data Collection

- Obtained café sales data as a single CSV file, `dirty_cafe_sales.csv`, containing 10,000 transaction records with columns: Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date.

Data Import

- Opened Power BI Desktop.
- Used **Get Data** → **Text/CSV** to select `dirty_cafe_sales.csv`.
- Loaded the file into the Power Query Editor for cleaning and transformation.

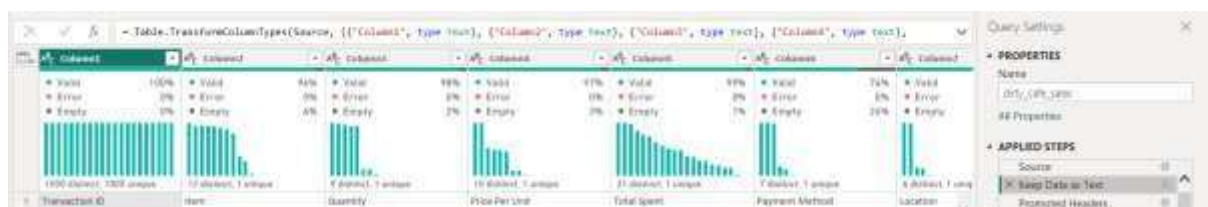
Data Cleaning and Transformation

Within the Power Query Editor, the following 22 steps were applied in sequence to progressively clean the raw dataset.

1. **Get Data** – Loaded `dirty_cafe_sales.csv` into the query editor. The raw café sales dataset was imported into Power BI using the Get Data option. The data contained different types of problems such as missing values, duplicate records, errors, and inconsistent entries. The dataset was then opened in Power Query Editor for cleaning.

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
TSM_1001271	Coffee	2	2.0	4.0	Credit Card	Takewave
TSM_1001272	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001273	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001274	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001275	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001276	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001277	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001278	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001279	Coffee	1	2.0	2.0	Credit Card	Takewave
TSM_1001280	Coffee	1	2.0	2.0	Credit Card	Takewave

2. Keep Data as Text – Set all columns to Text initially to avoid import errors. All the columns were initially changed to the **Text** data type. This was done to prevent Power Query from automatically converting incorrect or dirty values into unsuitable formats. It allowed the raw data to be checked safely before applying the correct data types.



3. Set Headers – Promoted the first row to become column headers (Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, Transaction Date). The first row of the dataset was promoted to become the **column headers**. This made it easier to identify.



4. Keep Columns as Text – Reconfirmed Text type as a safe baseline before deeper cleaning. The columns were kept as **Text** while the initial cleaning was performed. This helped avoid conversion errors caused by values such as ERROR, UNKNOWN, blanks, or incorrectly entered numbers.

5. Fix ERROR Values – Replaced all "ERROR" placeholder entries (present across Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date) with blanks. The dataset contained some **ERROR** entries in different fields. These values were replaced with blank values so that they would not interfere with later calculations and transformations.



6. Fix UNKNOWN Values – Replaced all "UNKNOWN" placeholder entries (present across the same set of columns) with blanks. Some records contained the value **UNKNOWN**, which did not provide useful information for analysis. These entries were replaced with blank values so they could be handled properly in the following cleaning steps.



7. Set Correct Data Types

After removing the obvious incorrect values, the appropriate data types were assigned to each column. Numerical fields were changed to **Number**, transaction dates were changed to **Date**, and descriptive fields were kept as **Text**.



- Quantity → Decimal Number (or Whole Number)
- Price Per Unit → Decimal Number
- Total Spent → Decimal Number
- Transaction Date → Date
- Transaction ID, Item, Payment Method, Location → Text

1.2	Quantity	1.2	Price Per Unit	1.2	Total Spent
1.2	Decimal Number	Valid	93%	Valid	95%
\$	Fixed decimal number	Error	0%	Error	0%

8. Fix Missing Total Spent – Some records had missing values in the **Total Spent** column. Where Quantity and Unit Price were available, the missing Total Spent was calculated Cthe missing Total Spent values (173 blanks) as Quantity × Price Per Unit.

New column name

Total Spent Fixed

Custom column formula ⓘ

```
= if [Total Spent] = null then [Quantity] * [Price Per Unit]
    else [Total Spent]
```

9. Fix Missing Quantity – Derived the missing Quantity values (138 blanks) from Total Spent ÷ Price Per Unit. Some transactions had missing **Quantity** values. When Total Spent and Unit Price were available, Quantity was calculated.

New column name

Quantity Fixed

Custom column formula ⓘ

```
= if [Quantity] = null then [Total Spent Fixed] / [Price Per Unit]
    else [Quantity]
```

10. Fix Missing Unit Price – Derived the missing Price Per Unit values (179 blanks) from Total Spent ÷ Quantity. Some records had missing **Unit Price** values. If Total Spent and Quantity were available, the Unit Price was calculated

New column name

Price Fixed

Custom column formula ⓘ

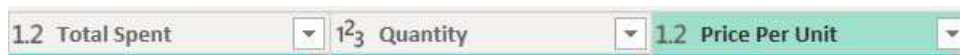
```
= if [Price Per Unit] = null then [Total Spent Fixed] /
    [Quantity Fixed] else [Price Per Unit]
```

11. Remove Old Columns – Removed the original columns that were superseded by the newly calculated fields. After creating corrected values, the unnecessary old or temporary columns were removed. This reduced confusion and kept only the required information in the final dataset

12. Fix Column Names – Renamed columns to clear, consistent names. The column names were checked and renamed where necessary to make them **clear, consistent, and easy to understand**. Proper column names also make future analysis easier.

13. Check Data Types – Re-verified and reset data types after the calculated columns were added. The data types were checked again after the calculations and transformations. This ensured that Quantity, Unit Price, Total Spent, and Transaction Date were stored in the correct formats.

Transaction ID	Item	Payment Method	Location	Transaction Date
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14. Fix Missing Items – Replaced the 333 blank Item values with a default placeholder label such as "Unknown Item." Some records had missing values in the **Item** column. These blank values were replaced with a suitable default label so that the records could still be identified and used during future analysis.

15. Fix Missing Payment Method – Replaced the 2,579 blank Payment Method values with a default placeholder label such as "Unknown." Missing values were found in the **Payment Method** column. These were handled by replacing the blanks with a suitable default value. This makes the payment information more consistent across the dataset.

Replace Values

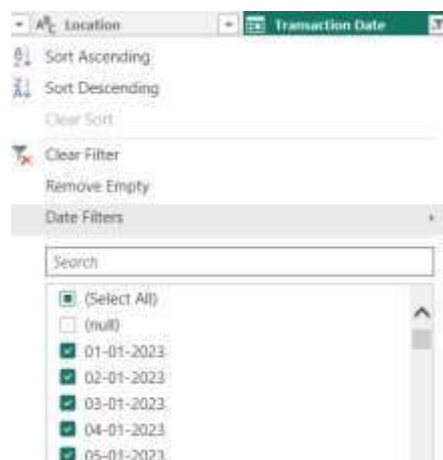
Replace one value with another in the selected columns.

Value To Find

Replace With

16. Fix Missing Location- Handled the 3,265 blank Location values, the largest gap in the dataset. Some café sales records did not contain **Location** information. These records were handled during the cleaning process so that incomplete location information would not affect future analysis.

17. Remove Missing Dates – Removed the 159 rows with a blank Transaction Date. Records with missing **Transaction Date** values were removed. The transaction date is important for understanding when a sale occurred, so records without a valid date could not be reliably used for date-based analysis.



18. Remove Duplicate Records – Checked for exact duplicate rows; none were found, confirming each Transaction ID was unique. Duplicate records were identified and

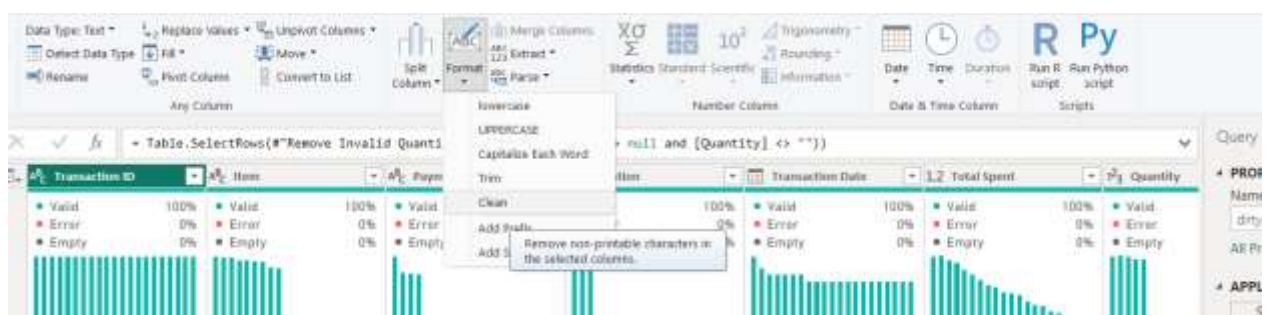
removed from the dataset. This prevents the same café transaction from being counted multiple times and helps maintain accurate sales information.



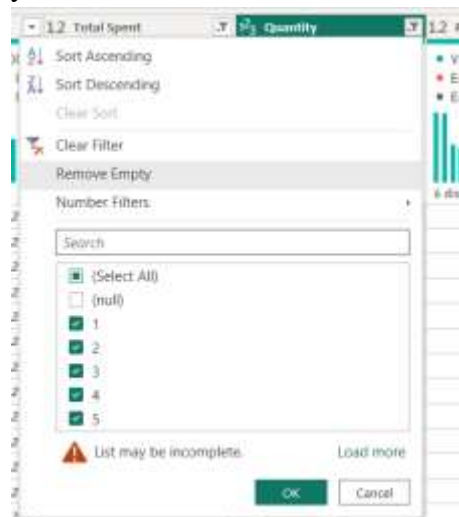
19. Fix Extra Spaces – Trimmed leading and trailing spaces from all text columns. Extra spaces before or after text values were removed using the **Trim** operation. For example, values such as " Coffee" and "Coffee " were cleaned so they could be treated as the same item.



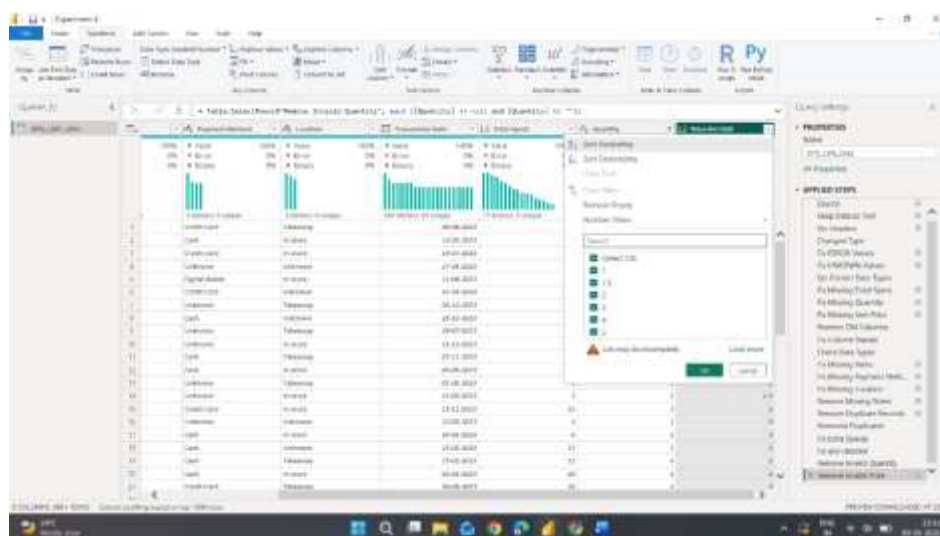
20 . Fixed and Cleaned – Removed non-printable and hidden characters from text fields. Unwanted or hidden characters in text fields were removed using the **Clean** operation. This improves the quality and consistency of text values and prevents unexpected differences between similar entries.



21. Remove Invalid Quantity – Removed rows containing invalid or negative Quantity values. The Quantity column was checked for invalid values such as **negative numbers or unsuitable entries**. Such records were removed because a normal café sale should have a valid positive quantity.



22 .Remove Invalid Price – Removed rows containing invalid or negative Price Per Unit values. The Unit Price column was checked for invalid or negative values. Records with unrealistic prices were removed to ensure that the final café sales dataset contained valid pricing information.



This sequence removed duplicate records, handled missing values (through calculation, default labels, or row removal), renamed columns appropriately, and converted data types where necessary.

Outcome

After applying all 22 transformation steps in Power Query, the dirty café sales dataset was successfully cleaned and organized. Errors, UNKNOWN values, missing values, duplicate records, incorrect data types, extra spaces, hidden characters, and invalid quantity and price values were identified and handled. Missing Total Spent, Quantity, and Unit Price values were also recalculated wherever possible.

As a result, the final dataset is more **accurate, consistent, and reliable** than the original raw data. The columns are properly named and formatted, and unnecessary or invalid records have been removed. The cleaned café sales dataset is now well-structured and ready for **further analysis in Power BI** and future business reporting.

Conclusion

This exercise demonstrated a complete data preparation and cleaning workflow using Power BI and Power Query. The raw café sales dataset was first imported into Power Query through the Get Data option and examined for different data quality problems. The dataset contained issues such as missing values, ERROR and UNKNOWN entries, duplicate records, incorrect data types, extra spaces, hidden characters, and invalid quantity and price values.

Power Query's step-by-step transformation features were then used to identify and correct these problems systematically. Missing Total Spent, Quantity, and Unit Price values were calculated wherever sufficient information was available. Unwanted and incorrect records were removed, duplicate transactions were eliminated, and text values were cleaned and standardized. The appropriate data types were also applied to ensure that numerical, text, and date fields were stored correctly.

Through this process, the original dirty café sales data was transformed into a clean, organized, and reliable dataset. The exercise helped demonstrate how data cleaning improves the accuracy and consistency of information before it is used for analysis. The final dataset is now properly structured and ready for further analysis in Power BI, helping ensure that future calculations, reports, and business decisions are based on accurate and trustworthy data.

Business Insights

- Clean data helps the café maintain **accurate sales records**.
- Missing **Total Spent, Quantity, and Unit Price** values can be calculated using the available sales information.
- Removing duplicate records prevents **sales from being counted more than once**.
- Correcting **ERROR and UNKNOWN values** improves the reliability of the data.

- Removing invalid or negative **Quantity and Unit Price** values ensures that the sales records are realistic.
- Cleaning **Item, Payment Method, and Location** information makes the data more consistent.
- Removing records with missing **Transaction Dates** helps maintain accurate sales records.
- Correct data types make it easier to perform **future sales analysis**.
- A clean dataset can help the café make better decisions about **sales, pricing, inventory, and customer preferences**.
- Overall, Power Query helps convert **dirty café sales data into clean and reliable data** ready for analysis.